IT 520 · Oct 14 activity: smaller, but still the same?

Name: _____

Work with a partner and use the reference box. Show your work: the steps are the answer.

Hex digits: 0–9, then A = 10, B = 11, C = 12, D = 13, E = 14, F = 15.
Two hex digits: `(left × 16) + right`. One hex digit is 4 bits.
RLE format: each run is one **record** = 1 count byte + 1 symbol byte. Each letter of input is 1 byte.
Invariant: after each record, decoding the records so far gives exactly the input consumed so far.

Measure compression
ratio = compressed size ÷ original size (below 1: smaller; above 1: larger)
savings = (original − compressed) ÷ original
Time
send time (s) = size (MB) ÷ speed (MB/s)
compressed route = encode + send + decode

Part A: Hex to decimal

1. A log shows the byte `2F`. What decimal value is it? Show `(left × 16) + right`.

2. Another byte is `C4`. What decimal value is it? Write it as 8 bits too (4 bits per hex digit).

3. An RLE count byte reads `0C`. (a) How many copies of its symbol does it stand for? (b) A count byte has 8 bits. How many different bit patterns can 8 bits hold? What is the largest unsigned value, written in hex?

Part B: Run-length encoding and its invariant

4. **Encode.** `AAAAAAAAAAAABBBBBBBC`. Write the records. Then give original size, compressed size, ratio and savings. Write the first record's count as a hex byte.

5. **Find the bad record.** An encoder turned `MMMMNNNNNNPP` into `(4,M),(5,N),(2,P)`. Check the invariant after each record:

After record	Input consumed so far	Decoded so far	Invariant holds?
1			
2			
3			

   Which record first breaks it, and what is the repair? Would checking only the final length have told you which record?

6. **Correct but bigger.** Encode `HELLO`. Does it decode exactly? Give the ratio. Why did it grow?

Part C: Choose a route

7. A 30 MB sensor log must arrive with **exact bytes**. Lossless compression makes it 6 MB (complete size). Encoding takes 2 s and decoding 1 s. MB ÷ MB/s gives seconds.

Route	At 3 MB/s	At 30 MB/s
Send original 30 MB		
Encode + send 6 MB + decode		
Your choice		

8. For each file, what recovery does the job require (exact bytes, or acceptable quality)? Which routes are allowed: lossless, lossy, or send unchanged?
   (a) a phone photo shrunk for a web page thumbnail (b) a program installer (c) a `.zip` file you are about to compress again